

Fig. 2: Netlist created for the given circuit

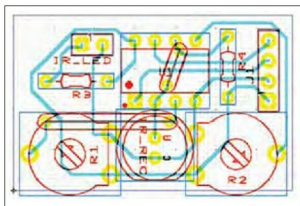


Fig. 3: Circuit board after routing of tracks

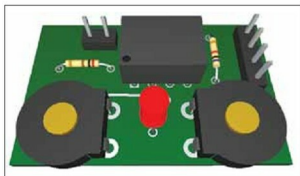


Fig. 4: 3-D view of the board

width and clearance to trace, which can be altered later also. Select units as millimetre for easy understanding and estimation at this stage. Track width should not be less than 0.5 mm if it is going to be manufactured by etching techniques. This generates a netlist file with a default double-sided board with all the components placed at its top edge. Once you reach this stage, you are ready to start board designing. This can be achieved by an alter-

native technique that starts from Ultiboard directly. Create a single- or double-sided board of the required size by selecting 'Board Wizard' main menu tools. Now place different parts from the database. Here, more concentration is given to the shape of the parts rather than the actual name of the part. Now create a different netlist using netlist editor from 'Tools' menu. Select the pins that are in a particular net and the board layer (copper top or bottom) where it will lie. You can also adjust widths of the track formed by any particular net.

With the netlist formed, you can do auto-placement of all the devices by clicking Autoplace from Autoroute menu. This places the components within the board. However, this hardly helps. You are required to place and orient the components on the board taking care of various constraints, like a serial port socket or a power supply socket should lie on the edge of the board.

Similarly, a crystal should be placed as close as possible to a microcontroller. Voltage regulators should be fitted on the edge so that these can be connected to the heat-sink if required. Some components like headers or female headers require to be placed for easy handling. You must optimise the placements for maximum efficiency of the system.

Resize the board after placing of all the components is finalised. After creating the network and adjusting

the board size, the layout may look as shown in Fig. 2. If the footprint for any part is not available in the database, the same can be created using component wizard from 'Tools' menu, by customising various dimensions like pin width, gap, drill diameter and solder pad dimensions. These dimensions may be found in the datasheet provided with the device or can be measured. The solder pad sizes should be kept to 0.5mm annular space if the components are to be soldered manually.

After creating the netlist and configuring various board parameters like track width and clearances, you may go for routing of the tracks. This can be done by selecting Autorouter from 'Autoroute' menu. Some tracks may not get auto-routed. These can be completed manually by placing tracks or fixing jumpers at some locations. The tracks are generated automatically depending on certain optimisation techniques for least track length, least number of through-hole connectors (Vias), etc. The tracks further require manual adjustments like straightening of tracks and placing of silk-screens on top and bottom. The final board looks as shown in Fig. 3. The same can be visualised in virtual by selecting View3D from 'Tools' menu.

Now that you have completed everything in soft, exchange this design with the PCB prototyping machines. One of the most widely used formats for this data exchange is GERBER RS-274X. Our design from NI Circuit Design Suite can be exported to this format by selecting 'Export' in 'Files' menu. Properties of Gerber files should be adjusted to print each layer that exists in the current design. Most important layers are copper top, copper bottom, board outline, drill and silk-screens. These files can be sent to different manufacturers for PCB prototyping.

The basic processes for other designs remain similar. Only the complexity of the design may increase. ●

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